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The Indian pharmaceutical industry is widely regarded as a successful example of industrial policy in action. Once marked by heavy import dependence, scarcity of drugs despite global availability, and extremely high prices, the sector has undergone a dramatic transformation. By the early 1990s, Indian pharma firms began proving their capability in the manufacture of generic medicines, forcing MNCs to enter into collaboration with them.

FROM SCARCITY TO A GLOBAL LEADER, INDIA TODAY:

- Exports medicines to 150+ countries
- Supplies 70 per cent of WHO's global vaccine needs
- Among the most affordable medicine suppliers globally
- Known as the 'Pharmacy of the World'

CHALLENGES TO SCALING UP PHARMA INNOVATION

- High risk and cost: Drug R&D typically takes 12 to 16 years and involves heavy capital investment
- Limited ecosystem: Only a few firms (e.g., Zydus Cadila, Wockhardt) have developed novel drugs
- Need: A broader network of deep-tech startups and innovation-driven firms



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A new leader in pharma

Manufacturing and exporting drugs in bulk, India is poised to become a prominent hub for pharmaceutical innovation

India's pharma sector is now deeply integrated into global value chains (GVCs), in both manufacturing and R&D. It accounts for four per cent of global exports of active pharmaceutical ingredients (APIs), which are the key components of finished dosage formulations, and is a major destination for offshore corporate pharmaceutical R&D.

Yet, given this robust global presence, a pertinent question arises: What relevance does the Make in India initiative hold for an industry already thriving? Let's take a closer look:

REDUCING IMPORT DEPENDENCE FOR RAW MATERIALS

A key vulnerability in the Indian

pharmaceutical sector is its excessive reliance on the imports of key starting materials (KSMs), drug intermediates (DIs), and certain APIs. While importing cheaper inputs and converting them into finished products in India makes economic sense, over-dependence on a single country poses strategic and supply chain risks. This has become particularly apparent in a world where supply chain dependencies can be weaponised.

To address this, the government introduced the first Production Linked Incentive (PLI) scheme for the pharma sector in 2020, aiming to reduce import dependence by incentivising local manufacturing of 41 products (KSMs, DIs, APIs) with high

import reliance. The results have been mixed. Imports of some products have declined, while others have seen an increase. Moreover, there are several products for which there were no or very few applicants, prompting the Department of Pharmaceuticals to reopen applications for 11 out of the 41 products as recently as May 2025.

A major constraint is a lack of cost competitiveness. Making domestic production viable will require substantial investments in process R&D and scale-building, which are both challenging in a context where 80 per cent of Indian API production comes from small or medium-sized enterprises. Revitalising the public sector through both enterprises and research institutions could be crucial, given their historic role in building India's generics capability.

FACILITATING THE TRANSITION TO AN INNOVATIVE INDUSTRY

Indian pharmaceutical firms account for nearly one-third of the country's industrial R&D expenditure. While R&D intensity (as a share of sales) has grown over time, this investment remains concentrated in a few large firms.

The second PLI scheme, launched in 2021, focuses on diversifying into high-value products, while the Scheme for Promotion of Research and Innovation in Pharma MedTech sector (2023) aims to shift the sector from cost-based to innovation-led growth.

To realise the goal of becoming a global pharma innovation hub, India must strengthen its deep-tech ecosystem and streamline regulations for clinical trials. Artificial intelligence is fast becoming a key tool in drug discovery, enabling faster timelines and lower costs. India must actively support AI-based pharma start-ups to remain competitive.

The Make in India initiative holds significant potential to redefine India's pharmaceutical strategy, not only by supporting domestic manufacturing but also by ensuring long-term resilience and fostering innovation. Reducing import dependence for critical inputs and enabling the growth of an innovative pharmaceutical ecosystem will require a coordinated policy framework, strategic public sector involvement, and sustained investment in R&D and infrastructure.

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